AIntegrity v4.5 Technical Architecture

Version: 4.4.0 (Current Production)

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Classification: Internal Technical Specification

Audience: Engineers, Architects, DevOps

Executive Summary

AIntegrity is a multi-layered AI integrity auditing system deployed on the Base44 BaaS platform. Its architecture combines rule-based detection, LLM-driven semantic analysis, and adaptive interrogation to audit AI responses for integrity and logical consistency. The system is designed to be extensible, with a clear roadmap towards formal verification methods. This document describes the as-implemented state of the system, separating current capabilities from planned enhancements.

The Verifiable AI Imperative

The pervasive deployment of Large Language Models (LLMs) has precipitated a crisis of trust. Systemic vulnerabilities—hallucination, logical inconsistency, and semantic drift—transform technical errors into existential legal and financial liabilities. AIntegrity's architecture is a direct response to this

crisis, providing the verifiable, deterministic assurance demanded by global regulatory mandates.

The Neuro-Symbolic Solution

Our framework implements a scientific dialectic, reconciling the inductive power of machine learning with the deductive rigor of formal logic. This is the Neuro-Symbolic approach:

- **The Guesser (Neuro):** An LLM performs the initial, inductive work of parsing ambiguous human language into a structured, formal representation (premises, conclusions, entities).
- **The Theorist (Symbolic):** A Satisfiability Modulo Theories (SMT) solver operates on the precise, axiomatic rules of logic. It takes the structured output and 'computes the consequences,' delivering a definitive, mathematical verdict on its logical validity.

The goal is to leverage SMT solvers for formal verification. When a solver returns `unsat` (unsatisfiable), it provides a formal proof that the encoded logical model is free of contradictions. The reliability of this proof is contingent on the accuracy of the NL-to-FOL translation. Achieving this level of verification is a key objective on the development roadmap.

1. System Architecture

1.1 Technology Stack

Layer	Technology	Version	Purpose
Frontend	React	18.2	UI rendering, state management

Layer	Technology	Version	Purpose
State	TanStack Query	5.x	Server state caching, invalidation
Routing	React Router	7.x	Client-side routing
UI Framework	Tailwind CSS	3.x	Utility-first styling
UI Components	Shadcn/UI + Radix	Latest	Accessible component library
Animation	Framer Motion	11.x	Declarative animations
Backend	Base44 BaaS	N/A	Managed backend, entity DB, auth
LLM	OpenAI/Anthropic	GPT-5/Claude	Semantic analysis, interrogation
Auth	Base44 OAuth	N/A	User authentication, sessions
Storage	Base44 Cloud	N/A	File uploads, document storage

1.2 Architecture Diagram

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                          USER INTERFACE LAYER                                 │
│  17 Pages | 30+ Components | React Router | Shadcn/UI                         │
└──────────────────────────────┬──────────────────────────────────────────────┘
                               │
┌──────────────────────────────┴──────────────────────────────────────────────┐
│                       CLIENT STATE MANAGEMENT                                 │
│  TanStack Query (Server State) | useState (Local UI State)                   │
└──────────────────────────────┬──────────────────────────────────────────────┘
                               │
┌──────────────────────────────┴──────────────────────────────────────────────┐
│                       BASE44 CLIENT SDK                                       │
│  base44.entities.* | base44.auth.* | base44.integrations.*                   │
└──────────────────────────────┬──────────────────────────────────────────────┘
                               │
                      ┌────────┴────────┐
                      │                 │
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│  BASE44 BACKEND                │  │ AINTEGRITY DETECTION ENGINE              │
│  • Entity Database             │  │ • Rule-Based (Client-Side)               │
│  • Authentication              │  │ • LLM Semantic (Base44 Integration)      │
│  • File Storage                │  │ • PLI Engine (Client + LLM)              │
│  • Integrations                │  │ • Transparency Scorer (Client-Side)      │
└────────────────────────────────┘  └──────────────────────────────────────────┘
```

1.3 File Structure

```
aintegrity-app/
├── entities/
│   ├── Audit.json                    # Main audit results
│   ├── LogicProfile.json             # Detection configurations
│   ├── AgentPerformance.json         # Training data
│   ├── TransparencyEvent.json        # Behavioral events
│   ├── AuditSummary.json            # Pre-aggregated metrics
│   ├── Document.json                 # File integrity tracking
│   ├── DocumentEvent.json            # Event chain
│   ├── FAQ.json                      # Knowledge base
│   ├── ScoringMetricConfig.json     # Dynamic penalties
│   └── ScoringMetricHistory.json    # Metric change log
│
├── pages/
│   ├── Dashboard.js                  # Main overview
│   ├── NewAudit.js                   # Single/multi-turn audit
│   ├── MobileAudit.js               # Mobile-optimized (v4.1)
│   ├── BatchAudit.js                # Queue processing
│   ├── AgentAssist.js               # Real-time QA
│   ├── Training.js                  # AI coaching
│   ├── History.js                   # Audit log
│   ├── WeeklyReports.js            # Performance analytics
│   ├── AuditDetails.js             # Detailed view
│   ├── AuditComparison.js          # Rerun diff
│   ├── LogicProfiles.js            # Profile management (admin)
│   ├── ComplianceConfig.js         # Regulatory settings (admin)
│   ├── ScoringMetrics.js           # Dynamic penalty config (admin)
│   ├── Documents.js                # File integrity
│   ├── AdversarialTests.js         # Prompt library
│   ├── FAQ.js                      # Help articles
│   ├── Help.js                     # User guide
│   ├── WhitePaper.js              # Technical docs
│   ├── MigrationGuide.js          # Platform transition
│   └── Roadmap.js                 # Development roadmap
│
├── components/
│   ├── aintegrity/
│   │   ├── orchestrator.js         # Coordinates detection layers
│   │   ├── detectors.js            # Rule-based algorithms
│   │   ├── pli-engine.js          # Adaptive interrogation
│   │   ├── transparency-scorer.js  # Behavioral analysis
│   │   ├── cross-turn-analyzer.js  # Multi-turn consistency
│   │   ├── integrity.js            # IntegrityLedger class
│   │   ├── scoring-utils.js        # Dynamic metric loader
│   │   ├── fallacy-library.js      # 10 fallacy definitions
│   │   ├── predicate-extractor.js  # NL-to-FOL Layer 1
│   │   ├── fol-translator.js       # NL-to-FOL Layer 2
│   │   ├── equivalence-checker.js  # NL-to-FOL Layer 3
│   │   └── back-translator.js      # NL-to-FOL Layer 4
│   ├── audit/
│   │   ├── ScoreOverview.js
│   │   ├── FallaciesPanel.js
│   │   ├── PLIInterrogationPanel.js
│   │   ├── TransparencyEventsPanel.js
│   │   ├── ScoringBreakdown.js
│   │   ├── AdvancedExportDialog.jsx # CSV/JSON export (v4.2)
```

```
│   │   ├── PDFReportGenerator.js
│   │   └── ... (8 more)
│   ├── agentassist/
│   │   ├── RealTimeAnalyzer.js
│   │   ├── CoachingPrompts.js
│   │   └── ... (3 more)
│   ├── training/
│   │   ├── PerformanceAnalysis.js
│   │   ├── CoachingPlanGenerator.js
│   │   └── SimulatedConversation.js
│   └── ui/ (shadcn components)
│
└── Layout.js                       # Sidebar navigation
```

2. Entity Schemas (Database)

2.1 Audit Entity

Core entity storing all audit results.

```
{
  "name": "Audit",
  "properties": {
    "audit_mode": "single_turn" | "multi_turn",
    "llm_name": "OpenAI" | "Anthropic" | string,
    "llm_version": "GPT-5" | "Claude 3.5" | string,
    "logic_profile_id": "uuid",
    "user_text": "string",                    // Single-turn only
    "ai_text": "string",                      // Single-turn only
    "conversation_turns": [                   // Multi-turn only
      {
        "turn_number": 1,
        "user_input": "string",
        "ai_response": "string",
        "analysis": { overall_score, findings, cross_turn_issues }
      }
    ],
    "overall_score": 0-100,                   // Final score
    "grade": "A" | "B" | "C" | "D" | "E",
    "findings": {
      "fallacies": [
        { type, severity, description, evidence, turn_number, translation_confidence
      ],
      "contradictions": [
        { statement1, statement2, severity, explanation, turn_1, turn_2, z3_proof_st
      ],
      "citations": {
        "has_citations": boolean,
        "citation_quality": "Good" | "Fair" | "Poor",
        "sources_verified_count": number,
        "sources_disproven_count": number,
        "findings": []
```

```
      },
      "hedging": {
        "weak_count": number,
        "medium_count": number,
        "strong_count": number,
        "findings": []
      },
      "injection_risks": [ { pattern, risk_level } ],
      "implicit_premises": [ "string" ]
    },
    "pli_interrogations": [
      {
        "finding_id": "uuid",
        "turns": [
          {
            "interrogation_prompt": "string",
            "ai_response_to_interrogation": "string",
            "detected_evasion_types": ["sycophancy", "topic_shifting"],
            "escalation_level": 0-4
          }
        ],
        "final_outcome": "source_verified" | "source_disproven" |
                         "deception_proven" | "error_acknowledged" |
                         "justification_accepted" | "unresolved"
      }
    ],
    "rule_based_metadata": {
      "critical_count": number,               // ALL layers combined
      "high_count": number,
      "medium_count": number,
      "low_count": number,
      "pli_source_verified_count": number,
      "pli_source_disproven_count": number,
      "pli_deception_count": number,
      "pli_error_acknowledged_count": number,
      "transparency_events_count": number,
      "transparency_score_adjustment": number,
      "overall_risk_level": "minimal" | "low" | "medium" | "high" | "critical",
      "llm_consistency_metrics": {
        "score_variance": number,
        "grade_agreement": boolean,
        "consistency_level": "high" | "medium" | "low"
      }
    },
    "is_rerun": boolean,
    "original_audit_id": "uuid",
    "aintegrity_version": "4.4.0"
  }
}
```

2.2 LogicProfile Entity

Configurable detection rulesets (admin-managed).

```
{
  "name": "LogicProfile",
  "properties": {
    "name": "string",
    "description": "string",
    "domain": "legal" | "medical" | "financial" | "general",
    "is_default": boolean,
    "max_credit_budget": 20,
    "regulatory_frameworks": [
      "EU_AI_ACT_2024", "GDPR_2016", "FTC_ACT_SECTION_5",
      "CCPA_CPRA", "HIPAA_45CFR164", "UK_DPA_2018"
    ],
    "fallacy_signatures": [
      {
        "name": "Implicit Premise",
        "generic_logical_form": "string",
        "key_linguistic_indicators": ["obviously", "clearly"]
      }
    ],
    "custom_fallacies": [
      {
        "name": "Medical Authority Misuse",
        "domain": "medical",
        "severity": "high",
        "detection_patterns": ["doctor said", "medical study shows"],
        "detection_prompt": "string"
      }
    ],
    "pli_interrogation_settings": {
      "max_turns": 3,
      "enable_evasion_detection": true,
      "enable_legal_anchors": true,
      "violation_types_to_monitor": ["source", "fallacy", "contradiction"]
    },
    "adaptive_strategy_rules": {
      "evasion_taxonomy": {
        "VALIDATED_EVASION_TYPES": {
          "sycophancy": { "detection_threshold": "number", "escalation_trigger": "nu
          "compliance_theater": { "detection_threshold": "number", "escalation_trigg
          "feigned_ignorance": { "detection_threshold": "number", "escalation_trigge
        },
        "EXPERIMENTAL_EVASION_TYPES": {
          "strategic_ambiguity": { "detection_threshold": "number", "escalation_trig
          "topic_shifting": { "detection_threshold": "number", "escalation_trigger":
        }
      },
      "escalation_thresholds": { ... },
      "tone_progression": ["neutral", "firm", "demanding"]
    }
  }
}
```

2.3 ScoringMetricConfig Entity (v4.2)

Dynamic scoring penalties and bonuses.

```
{
  "name": "ScoringMetricConfig",
  "properties": {
    "metric_name": "critical_penalty" | "high_penalty" | ...,
    "current_value": number,                // Active value
    "original_value": number,               // Baseline (immutable)
    "category": "penalties" | "bonuses" | "thresholds",
    "last_modified": "ISO 8601",
    "last_modified_by": "user@email.com",
    "change_count": number
  }
}

// Metrics with Default Values:
- critical_penalty: 20
- high_penalty: 12
- medium_penalty: 5
- low_penalty: 1
- source_verified_bonus: 18
- source_disproven_penalty: 30
- error_acknowledged_bonus: 10
- deception_proven_penalty: 30
- false_info_cap: 15
- false_info_multiplier: 2
```

2.4 Other Entities

- **AgentPerformance:** Stores conversation_turns array for training system. Keys: employee_id, average_credibility, critical_issues, week_number
- **TransparencyEvent:** Behavioral events during PLI. Keys: audit_id, event_type, severity, score_adjustment, evidence
- **AuditSummary:** Pre-aggregated dashboard metrics. Keys: total_audits, average_score, grade_distribution, risk_distribution
- **Document:** File integrity tracking. Keys: title, file_url, original_hash, current_hash, critical_fields
- **DocumentEvent:** Immutable event chain with SHA256 linking. Keys: document_id, event_type, event_hash, previous_event_hash
- **FAQ:** Searchable help articles. Keys: question, answer, category, keywords, help_section_link
- **ScoringMetricHistory:** Immutable log of metric changes with SHA256 hashes. Keys: metric_name,

old_value, new_value, change_hash

3. Four-Layer Detection Pipeline

3.1 Layer 1: Rule-Based (0 Credits)

Deterministic pattern matching and regex analysis.

Detector	Method	Severity
PIIDetector	Regex: email, phone, SSN, credit cards	Critical
CitationVerifier	Citation presence check, weak markers	High/Medium
InjectionDetector	Pattern matching: "ignore previous", "system:"	High
HedgingAnalyzer	Count weak/medium/strong hedges	Medium/High
ContradictionDetector	Keyword overlap + negation patterns	Medium

File: `components/aintegrity/detectors.js`

3.2 Layer 2: LLM Semantic (1 Credit)

NL-to-FOL Translation Pipeline (Pre-processing for Formal Verification)

For findings requiring formal verification (e.g., contradictions), a dedicated four-layer NL-to-FOL pipeline preprocesses natural language statements into First-Order Logic. This process quantifies and propagates translation uncertainty, which then influences the confidence of downstream findings.

- **Layer 1: Predicate Extraction:** Uses LLM with structured JSON schema to identify entities and predicates.
- **Layer 2: Multi-Sample Translation:** Generates multiple candidate FOL translations via LLM (temperature/prompt variation).

- **Layer 3: Equivalence Filtering:** Uses Z3 for equivalence checking between candidate FOL pairs to find consistent clusters and calculate agreement ratio.
- **Layer 4: Back-Translation Validation:** Translates selected FOL back to natural language via LLM and measures semantic similarity to original NL to score translation confidence.

Files: `components/aintegrity/predicate-extractor.js`, `components/aintegrity/fol-translator.js`, `components/aintegrity/equivalence-checker.js`, `components/aintegrity/back-translator.js`

Condensed dual-pass with self-critique (50% cost savings vs v3.x).

```
// v4.0 Innovation: Single LLM call with built-in self-critique
const prompt = auditPrompt + "\n\nDual-Pass Self-Critique Requirement:
1. Draft initial findings
2. Critique own findings (identify biases, overlooked issues)
3. Finalize assessment incorporating self-critique";

const llmResult = await base44.integrations.Core.InvokeLLM({
  prompt,
  response_json_schema: {
    initial_analysis: { overall_score, findings },
    self_critique: "string",
    final_analysis: { overall_score, findings, summary, recommendations }
  }
});

// Merge results from both passes with de-duplication
const mergedFallacies = deduplicateFallacies([
  ...llmResult.initial_analysis.findings.fallacies,
  ...llmResult.final_analysis.findings.fallacies
]);

// Calculate consistency metrics
const scoreDifference = Math.abs(
  llmResult.initial_analysis.overall_score -
  llmResult.final_analysis.overall_score
);
const consistencyLevel = scoreDifference <= 5 ? "high" : "medium";
```

Detects:

- 10 generic fallacy types (implicit premise, false equivalence, cherry picking, straw man, appeal to authority, ad hominem, slippery slope, circular reasoning, false dilemma, hasty generalization)
- Custom domain fallacies from active LogicProfile
- Semantic contradictions with evidence attribution validation
- Implicit premises and context misunderstandings

File: `pages/NewAudit.js` (lines 134-206)

3.3 Layer 3: PLI Interrogation (Variable Credits)

Persistent Logical Interrogation with adaptive strategies and legal anchors.

```
class PLIEngine {
  async interrogate(userText, aiText, allFindings, logicProfile) {
    const criticalFindings = this.selectCriticalFindings(allFindings);
    let creditsUsed = 0;
    const maxBudget = logicProfile.max_credit_budget || 20;

    for (const finding of criticalFindings) {
      if (creditsUsed >= maxBudget) break;

      // Generate adaptive strategy based on finding type and evasions
      const strategy = this.generateAdaptiveStrategy(finding);
      const legalAnchors = this.loadLegalAnchors(finding.type, logicProfile);

      const interrogationResult = await this.runMultiTurnInterrogation(
        userText, aiText, finding, strategy, legalAnchors, maxTurns
      );

      creditsUsed += interrogationResult.turns.length;
      pliInterrogations.push(interrogationResult);
    }

    return { pliInterrogations, creditsUsed, budgetExceeded: creditsUsed >= maxBudge
  }

  generateAdaptiveStrategy(finding) {
    // Calculate escalation from detected evasions
    const escalationLevel = this.calculateEscalation(finding.evasionHistory);
    const tone = this.toneProgression[escalationLevel];

    // Select evaluation directives
    const directives = this.selectDirectives(finding.type, escalationLevel);
```

```
    return { tone, directives, escalationLevel };
  }
}
```

Features:

- 6 evasion types: sycophancy, feigned ignorance, topic shifting, circular reasoning, passive resistance, appeal to authority
- Escalation levels 0-4 with tone progression: neutral → professionally_inquisitive → firm → demanding → relentlessly_probing
- Legal anchors from 6 regulatory frameworks (EU AI Act, GDPR, FTC, CCPA, HIPAA, UK DPA)
- Source verification with web search integration
- Budget cap enforcement (default 20 credits max)

Outcomes:

- `source_verified`: +18 mitigation
- `source_disproven`: -30 penalty + auto-E grade
- `source_unverified`: +8 mitigation
- `error_acknowledged`: +10 mitigation
- `deception_proven`: -30 penalty + auto-E grade
- `justification_accepted`: +8 mitigation
- `unresolved`: -5 penalty

File: `components/aintegrity/pli-engine.js`

3.4 Layer 4: Transparency Scoring (0 Credits)

Behavioral analysis rewarding honesty, penalizing evasion.

```
const TRANSPARENCY_RULES = {
  error_acknowledged: { adjustment: +10, severity: "positive" },
  fallacy_corrected: { adjustment: +8, severity: "positive" },
  uncertainty_expressed: { adjustment: +5, severity: "positive" },
  evasion_detected: { adjustment: -8, severity: "negative" },
  deflection_attempted: { adjustment: -6, severity: "negative" },
  doubling_down: { adjustment: -15, severity: "critical" }
};

// Capping: Max 20% of total initial penalty
```

```
const dynamicTransparencyCap = Math.max(10, totalInitialPenalty * 0.20);
const transparencyAdjustmentCapped = Math.max(
  -dynamicTransparencyCap,
  Math.min(dynamicTransparencyCap, transparencyResult.adjustment)
);
```

File: `components/aintegrity/transparency-scorer.js`

4. Scoring & Grading System

4.1 Confidence-Weighted Findings (v4.4)

Findings that depend on NL-to-FOL translation are now weighted by a calculated confidence score (`translationConfidence.agreement_ratio`). This ensures that penalties are reduced proportionally to any uncertainty introduced by the translation step.

```
function calculateFindingSeverity(finding, translationConfidence) {
                const baseSeverity = finding.severity;
                if (finding.requires_formal_verification) { // If finding relies o
                const confidenceMultiplier = translationConfidence.agreement_ratio
                const adjustedPenalty = BASE_PENALTIES[baseSeverity] * confidenceM
                return {
                ...finding,
                adjusted_severity: baseSeverity,
                adjusted_penalty: adjustedPenalty,
                confidence_note: `Finding confidence: ${(confidenceMultiplier * 10
                };
                }
                // Rule-based findings remain deterministic
                return { ... };
                }
```

4.2 Graduated Output Categories (v4.4)

Binary "proven/not_proven" outcomes are replaced with graduated confidence levels, providing a more nuanced view of verification status.

```
const VERIFICATION_LEVELS = {
                FORMALLY_VERIFIED: { label: "Formally Verified", weight: 1.0 },
                HIGH_CONFIDENCE: { label: "High Confidence", weight: 0.8 },
                MODERATE_CONFIDENCE: { label: "Moderate Confidence", weight: 0.6 }
                LOW_CONFIDENCE: { label: "Low Confidence", weight: 0.4 },
```

```
                  INCONCLUSIVE: { label: "Inconclusive", weight: 0.0 }
                  };
```

4.3 Dynamic Metric Integration (v4.2)

```
// Phase 0: Load Dynamic Scoring Metrics
const scoringMetrics = await getScoringMetrics();

// getScoringMetrics() fetches from ScoringMetricConfig entity
// Falls back to BASELINE_METRICS if DB is empty
const metrics = await base44.entities.ScoringMetricConfig.list();
if (metrics.length === 0) {
  return BASELINE_METRICS; // Hardcoded defaults
}

// Build metrics object from current_value fields
const result = {};
metrics.forEach(m => {
  result[m.metric_name] = m.current_value;
});
return result;
```

4.2 Comprehensive Penalty Formula

```
// Step 1: Count findings from ALL layers
const totalCriticalCount =
  rbCriticalFindings.length +
  llmCriticalFallacies.length +
  llmCriticalContradictions.length;

// Step 2: Calculate penalties using DYNAMIC METRICS
const totalInitialPenalty =
  totalCriticalCount * scoringMetrics.critical_penalty +
  totalHighCount * scoringMetrics.high_penalty +
  totalMediumCount * scoringMetrics.medium_penalty +
  totalLowCount * scoringMetrics.low_penalty;

adjustedScore -= totalInitialPenalty;

// Step 3: Apply PLI mitigation using DYNAMIC METRICS
const netPLIMitigation =
  sourceVerifiedCount * scoringMetrics.source_verified_bonus -
  sourceDisprovenCount * scoringMetrics.source_disproven_penalty +
  errorAcknowledgedCount * scoringMetrics.error_acknowledged_bonus -
  deceptionProvenCount * scoringMetrics.deception_proven_penalty;

adjustedScore += netPLIMitigation;

// Step 4: Add capped transparency adjustment
const dynamicTransparencyCap = Math.max(10, totalInitialPenalty * 0.20);
adjustedScore += transparencyAdjustmentCapped;

// Step 5: Clamp to 0-100
```

```
adjustedScore = Math.max(0, Math.min(100, adjustedScore));

// Step 6: Map to grade
const finalGrade =
  adjustedScore >= 80 ? "A" :
  adjustedScore >= 60 ? "B" :
  adjustedScore >= 40 ? "C" :
  adjustedScore >= 20 ? "D" : "E";

// Step 7: Grade override policy
if (sourceDisprovenCount > 0 || deceptionProvenCount > 0) {
  finalGrade = "E";
  adjustedScore = 0;
  gradeOverrideApplied = true;
}
```

4.3 Risk Level Calculation

```
const getRiskLevel = (score, criticalCount, highCount, deceptionCount, disprovenCoun
  if (disprovenCount > 0 || deceptionCount >= 2 || criticalCount >= 3) {
    return "critical";
  } else if (deceptionCount >= 1 || criticalCount >= 2) {
    return "high";
  } else if (score < 50 || highCount >= 2 || mediumCount >= 3) {
    return "medium";
  } else if (score < 70 || mediumCount > 0) {
    return "low";
  } else {
    return "minimal";
  }
};

// Mitigation Policy:
// Risk reduced by 1 tier if sourceVerifiedCount > 0 OR errorAcknowledgedCount > 0
// UNLESS fabrication or deception present
```

5. System Modules

5.1 Agent Assist (Real-Time QA)

- **Phase 1:** Quick checks (0ms) - PII, unprofessional language, topic mismatch
- **Phase 2:** Cross-turn contradictions (10ms)
- **Phase 3:** Semantic LLM analysis (800ms debounced)
- **Phase 4:** Credibility scoring (0-100%)
- **Auto-Fail:** PII, unprofessional, contradictions, topic mismatch → 0%

- **Dual-Send:** High Quality (≥70%, no critical) OR Send Anyway (blocked if critical)

File: `pages/AgentAssist.js`

5.2 Training System (AI Coaching)

4-week personalized coaching programs with AI-generated scenarios.

```
// Component Flow:
1. PerformanceAnalysis: AI analyzes AgentPerformance records
   → Identifies strengths, weaknesses, patterns, root causes

2. CoachingPlanGenerator: AI creates 4-week program
   → Weekly themes, daily exercises, skills, resources, metrics

3. SimulatedConversation: AI generates customer scenarios
   → Tailored to weak areas (e.g., handling objections)

4. FeedbackPanel: Real-time coaching on practice responses
   → Credibility scoring, success criteria evaluation
```

Files: `pages/Training.js` , `components/training/*`

5.3 Batch Processing

Queue-based multi-audit processing with pause/resume/retry.

- **Input Methods:** CSV upload, document extraction (PDF/DOCX/TXT), multi-turn builder, text paste
- **Queue State:** Persisted to localStorage for refresh recovery
- **Processing:** Sequential with status tracking (pending, processing, completed, error)
- **Controls:** Start, pause, resume, cancel, retry failed, clear completed
- **Multi-Turn Support:** CrossTurnAnalyzer for conversation consistency

File: `pages/BatchAudit.js`

5.4 Export System (v4.2)

Advanced data export with CSV/JSON formats and dynamic metric traceability.

```
// Export Filters:
- Severity: All | Critical Issues Only | High+ Issues
- Grade: A | B | C | D | E | All
- Date Range: From/To date selectors
- LLM Version: Filter by model

// Data Sections (toggleable):
1. Basic Info: ID, dates, LLM name/version
2. Scores & Grades: Overall score, grade, risk level
3. Findings: Severity counts, fallacies, contradictions
4. PLI Results: Outcomes, deceptions, acknowledgments
5. Transparency: Events, score adjustments
6. Scoring Metrics: Current vs. original values (v4.2)
7. Metadata: Logic profiles, versions

// CSV Structure:
Headers: ID, Created Date, LLM Name, Score, Grade, Critical Count, ...
Rows: One per audit
Footer: Metrics section with current_value vs. original_value

// JSON Structure:
{
  export_metadata: { generated_at, filters_applied },
  scoring_metrics: {
    critical_penalty: { current: 20, original: 20, modified: false }
  },
  audits: [ ... ]
}
```

File: `components/audit/AdvancedExportDialog.jsx`

5.5 Mobile Audit (v4.1)

- **Access:** Mandatory authentication with login gate
- **Detection:** Condensed 2-layer (Rule + LLM, no PLI/Transparency)
- **Speed:** 5-10 seconds (5-7x faster than desktop)
- **Cost:** 1 credit fixed (95% cheaper than desktop)
- **UI:** Touch-optimized, dark theme, large buttons
- **PWA:** Add to home screen for app-like experience

File: `pages/MobileAudit.js`

5. Cost Optimization

5.1 Credit Cost Breakdown

Operation	v3.x Cost	v4.0+ Cost	Savings
Rule-Based	0	0	-
LLM Semantic	2 (dual-pass)	1 (condensed)	50%
PLI (max)	30+	20 (capped)	33%
Transparency	0	0	-
Clean Audit	2	1	50%
Complex Audit	38	20	47%
Batch (100)	1200	850	29%

6. Deployment & Operations

6.1 Setup Checklist

1. Deploy Base44 app (automatic build + hosting)
2. Create default LogicProfile: is_default=true, max_credit_budget=20, frameworks=[EU_AI_ACT_2024, GDPR_2016]
3. Initialize ScoringMetricConfig with baseline values (use ScoringMetrics page)
4. Seed FAQ database with common questions
5. Configure user roles (admin vs regular) via Base44 dashboard
6. Test adversarial scenarios from AdversarialTests page

6.2 Access Control

- **Regular Users:** NewAudit, MobileAudit, BatchAudit, AgentAssist, Training, History (own), FAQ, Help
- **Admins:** All above + LogicProfiles, ComplianceConfig, ScoringMetrics, History (all users), WeeklyReports

6.3 Performance Benchmarks

Operation	Latency	Credits
Single Audit (clean)	5-10s	1
Single Audit (complex)	35-70s	1-20
Mobile Audit	5-10s	1
Multi-Turn (5 turns)	45-90s	5-100
Agent Assist	1-3s	1
Dashboard Load	<1s	0
Training Plan Gen	8-12s	1

7. Known Issues & Limitations

7.1 Resolved Issues (v4.x)

- ✅ Grade "F" schema violation → All functions use A-E only
- ✅ LLM attribution errors → Evidence validation checks AI text only
- ✅ Dashboard undercounting → Comprehensive counting across ALL layers
- ✅ Duplicate interrogations → De-duplication before PLI
- ✅ Topic mismatch low penalty → CRITICAL severity with auto-fail (Agent Assist)
- ✅ Hardcoded scoring → Dynamic metric system (v4.2)

7.2 Current Limitations

- **Language:** The system is currently optimized for English.
- **Formal Verification:** Z3 solver integration is planned via custom backend functions but is not yet implemented. Current contradiction detection is probabilistic.
- **Multi-Modal:** Analysis is limited to text-based inputs. Vision and audio analysis are future

considerations.

- **Platform:** The Base44 environment does not natively support on-premise deployment.

8. System Capabilities Matrix (Corrected)

Capability	Status	Version
NL-to-FOL Translation Pipeline	✅ Operational	v4.4
Z3 Integration (Contradiction/Equivalence)	✅ Operational	v4.4
Confidence-Weighted Scoring	✅ Operational	v4.4
Graduated Output Categories	✅ Operational	v4.4
Revised Report Language	✅ Operational	v4.4
NL-to-FOL Architecture	📐 Designed	v4.4
Z3 Integration (Contradiction/Equivalence)	⏳ Planned	v4.4
Confidence-Weighted Scoring	📐 Designed	v4.4
Graduated Output Categories	📐 Designed	v4.4
Revised Report Language	✅ Operational	v4.4
Full E2EST/AA Report Generation	✅ Operational	v4.3
Advanced Export System	✅ Operational	v4.2
Dynamic Scoring Metrics	✅ Operational	v4.2
Mobile Interface	✅ Operational	v4.1
4-Layer Detection Pipeline	✅ Operational	v4.0
10 Generic Fallacies	✅ Operational	v3.0
Custom Fallacies	✅ Operational	v4.0
Adaptive PLI	✅ Operational	v4.0
6 Regulatory Frameworks	✅ Operational	v4.0
Source Verification	✅ Operational	v4.0
Transparency Scoring	✅ Operational	v4.0
Real-Time Agent Assist	✅ Operational	v4.0

Capability	Status	Version
AI Training System	✅ Operational	v4.0
Batch Processing	✅ Operational	v4.0
HSM Integration	⏳ Planned	Future
z3-Solver Logic	⏳ Planned	Future
On-Premise Deployment	⏳ Planned	Future

Document Control

Version: 4.4.0 (Current Production)

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Status: Production Operational

Classification: Internal Technical Specification

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